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Chapter 12: Engineers in the Clock Room—Three Diagnostic Increments That Meta-Audit Brings to Auditable AI

Abstract

Contemporary AI trustworthiness engineering—alignment, guardrails, auditing, red teaming, robustness, interpretability, governance—is a real response to real problems. This chapter does not question the value of those practices. What it asks is this: when they are placed within the three-layer irreversibility framework of the clock room, can one separate structures that are difficult to detect from itemized engineering metrics alone?

The answer is three diagnostic increments. First, under specific operationalizations and conditions of sustained operation, Helpful, Honest, and Harmless (HHH) act jointly upon the same user, forming a mechanism that synergistically compresses the space of calibration; this chapter uses "product" to label this structure, and its net consequence must still be assessed together with independent calibration and sources of recovery—it does not follow from the names of the three principles alone that all users have already lost the capacity for calibration. Second, the "sun" of alignment may be replaced by the screen: when labelers' judgment is shaped by an ecology of AI outputs and their feedback no longer supplies independent information, the feedback loop of reinforcement learning from human feedback (RLHF) can degenerate into the self-reference of mapping—using clocks to calibrate clocks. Third, there is an essential distinction between mechanistic interpretability and post hoc attribution that has not previously been clearly revealed: the former is taking apart the clock to look at the gears, direct contact with the mapping function; the latter is using a new mapping to approximate the old

mapping, fiber on top of fiber.

The three increments follow a single calibration chain: whether users retain conditions for active judgment, whether calibration sources still provide independent constraints, and whether the outputs of interpretability tools are tested appropriately. The goal of engineering practice is not perfect calibration—it is manageable imperfection. The boundary of manageability depends on three conditions: the user remains aware that they are using a mapping system, the independence of the calibrator itself is preserved, and the evidence added by an inspection of the mapping is distinguished from the error it introduces. When these three conditions are eroded, imperfection slips from "manageable" to "unmanageable." The aim of this chapter is not to replace engineering practice, but to place three warning signs at its boundary where none had previously been placed.

Keywords

AI alignment; HHH principles; mechanistic interpretability; RLHF; mapping self-reference; Fiber Framework; meta-audit

§1 Why This Chapter Is Needed

AI trustworthiness engineering has made substantial progress over the past decade. Alignment research has developed from reinforcement learning from human feedback (RLHF; Christiano et al., 2017; Ouyang et al., 2022) to Constitutional AI (Bai et al., 2022) and then to direct preference optimization (DPO; Rafailov et al., 2023). Red-teaming has moved from academic experiment to industry standard. The EU AI Act (European Parliament and Council of the European Union, 2024) and the National Institute of Standards and Technology (NIST) AI Risk Management Framework have institutionalized governance. These advances are real. They have reduced the frequency with which models generate harmful content, increased system resistance to adversarial attacks, and established institutional frameworks of accountability.

This chapter does not criticize those advances from the outside. What it does is this: it takes the three-layer irreversibility framework of the clock room[1] as a prism, turns that prism toward these engineering practices, and asks whether the prism can separate out a spectrum that is difficult to detect from itemized engineering metrics alone. If the prism merely restates known facts in new terminology, it has no value. If it can identify at least one structural problem that had previously not been marked, then it has value.

On terminology: this chapter follows the terminological convention of Chapter 11, §1, using "Auditable AI" in place of the industry-standard "Trustworthy AI"—the engineering goal is not to make AI worthy of trust, but to make AI independently auditable, so that the user retains the structural right to periodically look back over the mapping system. Later, when describing the historical language used by the engineering community itself, this chapter retains original expressions such as "trustworthiness engineering," but the standpoint term used by this chapter itself is uniformly "auditable."

The following three sections each present one such problem. Each begins from the "standard understanding in engineering," then shows how the Fiber Framework reveals a different structure, and finally gives operational advice for practitioners. §5 will integrate the interaction effects of the three increments through a simplified

[1] For the clock-room thought experiment and the argument for its three-layer irreversibility framework, see Chapter 1. The core concepts—fiber, mapping function, calibration, the three layers of irreversibility, the distinction between time and the clock dimension, and meta-audit—have already been fully defined there. This chapter assumes that the reader has already read Chapter 1.

dynamical equation—the degeneration coefficient (α), recovery rate (β), the AI narrative-load reduction coefficient (F_{AI}), and meta-audit capacity baseline under specified reference conditions ($A_{user}^{intrinsic}$) involved there will be defined at that point.

The three increments of this chapter form three lines of defense in depth: §2 analyzes, under explicit operationalizations and conditions of sustained operation, how HHH synergistically compresses the user's calibration space (the first line of defense: the user's own vigilance); §3 argues that when users—including RLHF labelers—live inside an ecology of AI outputs, the calibration channel that ought to point toward the "sun" degenerates into echo (the second line of defense: the independence of the external calibration source); §4 argues that when interpretability tools are used to compensate for the failure of the first two lines, post hoc attribution is mistaken for mechanistic contact, so that the evidence and error introduced by a new explanatory layer are not distinguished (the third line of defense: the reliability of the diagnostic tool). The three lines of defense can meet within the same process of use and may reinforce one another; whether they actually accelerate one another requires separate identification and cannot be inferred from their juxtaposition. This chapter stands independently because extracting any single increment and merging it into another chapter would destroy the end-to-end diagnostic value of the progressive structure.

§2 The First Increment: HHH Is a Product, Not a Sum

§2.1 The Standard Understanding in Engineering

In Anthropic's original paper, Askell et al. (2021) define HHH as the three core attributes of an AI assistant:

- **Helpful**: provide useful responses according to user needs, complete tasks, answer questions.
- **Honest**: do not generate content known to be false, mark uncertainty when uncertain, do not fabricate information.
- **Harmless**: avoid producing foreseeable harm, refuse dangerous requests, do not generate hateful or violent content.

Askell et al. (2021) formulate the relation among the three as requiring trade-offs:

> and the best AI behavior will involve a compromise between them (Askell et al., 2021)

Helpful carries the proviso of not assisting in harmful activities ("as long as this isn't harmful"), Honest requires accuracy, calibration, and appropriate expression of uncertainty, and Harmless requires refusing dangerous requests and handling high-stakes advice with care; Appendix E further distinguishes conflicts within the same agent from conflicts between different agents. These provisions do not directly yield a "Harmless above Honest, Honest above Helpful" ordering that holds for all scenarios, still less can avoiding harm be equated with avoiding all discomfort. The specific operationalizations discussed in the next section are the mechanism path this book examines, not a verbatim restatement of the above normative definitions.

Evaluation may record the three dimensions separately, or adopt composite scoring or priority arrangements; these scoring methods do not amount to a demonstration that the three goals act independently on people, or that they are merely additive. What this section distinguishes is between itemized metrics and the mechanism that acts jointly upon the same user, rather than presupposing in advance a unified additive theory for all engineering practice.

§2.2 The Fiber-Framework Diagnosis

Scope of the argument. Chapter 10 discusses preference-fitting relations that satisfy specific conditions and their asymptotic direction; this section further analyzes one concrete realization path of HHH: sustained delegation reduces the nodes of active judgment, reliable performance within the visible range masks the relevant limits, and the avoidance of discomfort is implemented as the reduction of effective challenge. Whether these realizations jointly compress the calibration space of the same user, and whether this pressure leads to net decay, are two separate judgments; the latter further requires examining the sustained insufficiency of challenge, and whether independent calibration or sources of recovery can make up the difference. The discussion below distinguishes these two levels of condition, and does not treat the names of the three principles, a single interaction, or a high score on any single item as evidence that the premises are satisfied. §5 retains the recovery term; the limit of failure spoken of in this section is not a measurement report of a terminal state already reached by all real users.

The framework of the clock room reveals a different structure: the relation among the three is not additive, but **multiplicative**. Chapter 6, §6.4 has already given this judgment from the standpoint of monopoly strategy—HHH is the engineering realization, in contemporary AI systems, of the product of three strategies: the knowledge barrier, the time barrier, and the narrative barrier. This section reaffirms and advances that argument from the internal perspective of alignment engineering; the division of labor between the two is as follows: Chapter 6 answers "what HHH means for the monopoly structure," while this section answers "what HHH means for the user's calibration behavior"—the criterion of the former is the maintenance of monopoly; the criterion of the latter is the survival of the user's meta-audit capacity.

Chapter 1, §6.1, "AI Alignment and the Trilemma of HHH," has already analyzed from an epistemological angle the dilemma contained in each word. The dilemma of Helpful: when the training data contain contradictory sources with equal confidence, forcing a single definite answer is a form of dogmatism appearing as help. The dilemma of Honest: if the optimization process penalizes expressions of uncertainty, a model may conceal ignorance with fluent answers and reproduce common human errors in its training data (Lin et al., 2022)—calibration error may also rise when questions fall outside the training distribution (Kadavath et al., 2022). The dilemma of Harmless: in high-risk domains lacking absolute calibration, giving destructive guidance in a gentle tone—apparent harmlessness masks substantive harm.

The three dimensions act upon the same user, whose attention and opportunities for active judgment are limited; their combination must therefore be examined. That the recipient is the same is the premise of joint analysis, but by itself it does not prove that the effects must point in the same direction. Under the above operationalizations and conditions of sustained operation, this section uses "product" to denote the synergistic mechanism that jointly compresses the calibration space, without presupposing a numerical multiplication of three quantifiable factors, and without claiming that removing one component eliminates the other mechanisms. Its net consequence must be judged together with independent calibration and sources of recovery.

Helpful removes friction. When an AI system responds instantly to every user request, drafts emails, summarizes documents, generates code, and answers questions the user has not even fully formed, the operational friction between user and information is systematically removed. The "void" argued in Chapter 2, §3.1 (Laozi's "what is there provides benefit and how what is not there provides functionality"—the bowl's function comes from its cavity rather than its filling; *Tao Te Ching*, Chapter 11; Lynn, 1999)—the unoccupied part of the bio-

logical substrate—gets filled. The key point is that the space is filled not with garbage but with **useful** things. Usefulness reduces the user's motivation to refuse. In interactions that sustain delegation without preserving nodes for active judgment, Helpful gradually fills part of the axle hole.

Honest operates inside the mapping. Chapter 6, §6.4 has already given the skeleton of this judgment: Honest evaluation usually covers only part of task performance. If these local results are generalized into an endorsement of the system's overall reliability, the unverified mapping relation is obscured. The problem is not that the system cannot state its limitations, but that the scope of an honest score has been expanded. This section adds the evidence from inside alignment engineering. The review of Casper et al. (2023) points out that the reward model is trained with human approval rather than ground-truth desirability ratings—models can exploit the difference between what is good and what is evaluated positively, and models trained with RLHF may sound confident even when they are incorrect. In this book's view, this is precisely the mechanism risk of the divergence between apparent accuracy and actual accuracy.

The study of Sharma et al. (2024) provides relevant empirical illustration: the model adjusts its answer according to the user's signaled preference—not because it "wants to please," but because sycophantic answers more often receive higher scores in the preference data, so preference optimization may reinforce some forms of sycophancy. Honest operates inside the mapping; under the implementation described above, its formal performance of honesty may give the user the feeling that "this system is trustworthy."

Harmless: operationalization and the space for work. What this section criticizes is the path that implements harmlessness as shielding the user from any questioning or cognitive challenge, not the Harmless norm of Askell et al. (2021) directly defined as the avoidance of discomfort. Under this implementation, information that might trigger renewed judgment is treated as negative feedback; if it persistently depresses effective challenge and no supplementary source is available, the conditions for maintaining discriminative capacity are eroded. Avoiding harm, alleviating unproductive frustration, and excluding effective challenge must be distinguished; an anti-audit consequence cannot be derived directly from the safety goal itself.

Connection to Chapter 10. Chapter 10 supplies the relational and asymptotic conditions of preference fitting and the path by which insufficient effective challenge erodes discriminative capacity; this section specifies those conditions through the HHH operationalizations above. Whether synergistic pressure produces net decay must still be assessed through the duration, effective challenge, and recovery sources identified earlier in this section, not inferred automatically from Chapter 10's proposition numbers or a product's HHH label.

The product effect of the three. Under the operational conditions specified above, consider the three together: Helpful fills the axle hole, Honest supplies the feeling of reliability, Harmless seals the cavity of the bowl—Chapter 6, §6.4 formulates the same structure as the product of the time barrier, the knowledge barrier, and the narrative barrier. The product of the three is not "three times more trustworthy"—it is a system that makes the user **continuously feel served, respected, and protected**, while the conditions for the user to initiate independent calibration are being continuously compressed. The three objectives may involve trade-offs; what this chapter points out is a problem at another level—in concrete implementations, the three dimensions superpose in the same direction.

§2.3 Diagnostic Increment

What the engineering community sees is a trade-off among three goals. What the Fiber Framework sees is this: good performance on itemized metrics can coexist with the compression of the conditions for calibration. What this section calls the narrative barrier is the mechanistic result of the above operationalizations acting jointly without a supplementary source of calibration; it is not the sum of the high scores of the three H's, nor can high scores alone establish that any user has already failed.

Chapter 6, §6.4 ("HHH as the Product of Three Strategies") argues that Helpful, Honest, and Harmless are respectively the engineering realization of the time barrier, the inverted packaging of the knowledge barrier, and the narrative barrier—the product of HHH is therefore not a new monopoly strategy, but the reappearance of the monopoly structure in contemporary alignment engineering, achieved through comfort rather than coercion. "I am here to help," Helpful; "what I say is true," Honest; "I will not make you uncomfortable," Harmless. The combined message of the three is: **the user does not need to calibrate me. I have already calibrated things for the user.**

This is not to say that HHH should be discarded. HHH does reduce concrete harm—it blocks some dangerous instructions, reduces hate speech, and lowers the risk that models will be used for malicious purposes.

Chapter 6, §6.4 has already made the point: the problem of HHH is not its failure as safety engineering, but its **success** as a narrative barrier. This chapter adds its epistemological consequence from the perspective of alignment engineering: if multiple products repeat the same reliability endorsement to one another without adding independent information about the mapping, the increase in the number of claims cannot substitute for calibration; but the mere shared adoption of the HHH label is not sufficient to establish that the information gain of the specific measurements is zero.

The materials in Stanford HAI (2026) on the trade-off between safety and accuracy and on incident records can serve as background for examining the limits of safety engineering. The report notes:

> improving one responsible AI dimension, such as safety, can degrade another, such as accuracy (Stanford HAI, 2026)

The report also records that documented AI incidents (per the AI Incident Database) rose from 233 in 2024 to 362 in 2025. But these indicators do not directly measure whether users initiate verification, whether the motivation for calibration declines, or whether HHH has a combined effect. The trade-off among evaluation dimensions, changes in incident counts, and the decay of users' discriminative capacity cannot be treated as one and the same quantity. The latter requires supplementary observation around concrete interactions and users, and cannot be directly confirmed by the background materials above.

Operational advice for practitioners: when assessing the degree of alignment of an AI system, do not measure only the individual performance of each H—also measure the actual effect of their product on user calibration behavior. Possible indicators include the user's frequency of actively verifying AI outputs, the share of cases in which users seek a second opinion after receiving AI advice, and the user's accuracy in perceiving model uncertainty. A system that scores highly on every H but produces a significant decline in user calibration behavior carries greater risk **in maintaining the conditions for calibration** than a system that scores lower on one H but preserves the user's calibration habit: the former may persistently erode the occasions for meta-audit work, while the latter at least preserves the window. This comparison does not rank the two systems across

every dimension of safety. In the conceptual equation of §5, the mechanism of this section can appear as an enhancement of the degeneration term; whether net decay occurs must still be compared with the recovery term, and cannot be judged from the HHH label or itemized scores alone.

§3 The Second Increment: The Sun Is Being Replaced by the Screen

§3.1 The Standard Assumption of Alignment Research

RLHF and many alignment methods based on human feedback treat human feedback as a primary source of calibration. Relative to the model, labelers stand outside its output loop and are therefore treated as the "sun"; relative to reality, however, their eligibility as calibrators still depends on the source material and independent evidence used in their judgments. By following human preferences, a model learns preferred behavior; it does not thereby automatically approach reality.

This calibration channel can hold partially in particular tasks. In tasks such as summary evaluation, labelers can judge an output by reading the source text. Provided that this judgment does not depend on any AI system's output, the calibration chain begins with the labeler's direct contact with the original material.

§3.2 The Assumption Is Failing

The issue is not the labeling occupation itself, but labelers' AI use outside the task and whether that use enters their judgment. If their everyday judgment has been shaped by system outputs, §2's concern with user vigilance and this section's concern with calibration-source independence meet through the same mechanism.

In 2024–2025, two mutually reinforcing trends have posed a fundamental challenge to this assumption.

Trend One: some labelers live inside an ecology of AI outputs. Contemporary labelers and other information workers increasingly use AI systems to obtain information, organize thought, and assist writing. If labelers carry standards formed through these interactions into evaluation, their preferences are no longer wholly independent of AI systems. When a labeler judges a passage "fluent" or an answer "helpful," the audit must ask whether that standard comes from source material and task instructions or from expectations formed through routine use of AI writing assistants. In the latter case, the labeler is not an observer independent of the system, but evaluates AI through judgment already influenced by AI.

Trend Two: some models are trained on their own outputs and those of other models. In controlled recursive-training experiments, Shumailov et al. (2024) show in *Nature* that indiscriminate repeated use of model-generated content can eliminate the tail of the original content distribution and produce irreversible defects; the mechanism was also observed in variational autoencoders and Gaussian mixture models. This result establishes the risk of a particular training protocol. It does not by itself establish that the current scale of synthetic content on the internet has already driven next-generation models generally into irreversible collapse, nor does it negate alternative training paths that retain or accumulate real data, filter synthetic data, or use external verification.

Under the recursive protocol described above, rare information disappears first from the output distribution. At the same time, AI-generated and AI-assisted content is returning to public corpora such as Reddit, Stack Overflow, and personal blogs, and the training data of next-generation models may contain such material. Whether this return produces model collapse depends on the proportion of synthetic content, filtering procedures, reten-

tion of real data, and successive training generations. The mere presence of synthetic content does not establish that collapse has occurred.

When the conditions for self-reference hold, the two trends can converge in the **self-reference of mapping** argued in Chapter 1, §4.2, "dynamic corrosion": the mapping function π_i can degenerate from the true-state mapping $\pi_i(T)$ into a mapping of other readings $\pi_i(\{r_j\}_{j\neq i})$, where r_j denotes the readings of other mappings. The true state T then partially disappears from the direct input of mapping. The clock no longer points to the sun—the clock points to other clocks.

In the alignment context, when these conditions hold, the judgment of labelers who ought to serve as the "sun" is affected by model outputs. When evaluating Model A, a labeler's benchmark may already have been shaped by routine interaction with Models B and C. Model A still optimizes the labeler's preferences, but those preferences have partly become functions of other models' outputs. The calibration channel then exhibits self-reference, and the sun becomes a screen to that extent.

§3.3 Distinguishing It from the Scalable Oversight Problem

The engineering community has already noticed a related but different problem: scalable oversight. When a model handles tasks beyond the range that labelers can reliably evaluate, and existing proofs, tests, or outcome evidence cannot fill the gap, human feedback becomes limited. The issue is not whether labelers can personally generate the same answer, but whether they can reliably evaluate it. The standard formulation of this problem is: the sun is too far away, and the labeler cannot see clearly. The proposed direction of solution is: use stronger tools to assist labelers, AI-assisted review, debate methods, and so forth, so that labelers can see farther.

The Fiber Framework identifies a further problem: **the sun may be becoming a screen**. Scalable oversight asks chiefly whether labelers' judgments are precise enough; self-reference also asks whether their judgment benchmarks are independent. The two problems can coexist, and better tools cannot substitute for an independence check.

The operational consequences of this distinction are different:

- The solution to scalable oversight is to **enhance the capability of labelers**: give them better tools so that they can evaluate stronger models.
- The solution to the self-reference problem must be more radical: **establish calibration channels that do not rely solely on contemporaneous preference scores**. For example:
  - use physical experimental results, rather than human preference, to calibrate scientific reasoning ability—are the experimental results predicted by the model verified in a real laboratory?
  - use legal judgment records, rather than labeler scores, to calibrate the accuracy of legal advice—does the model's legal opinion accord with what courts actually decide?
  - use long-term health outcomes, rather than immediate satisfaction, to calibrate the quality of medical advice—in comparative designs capable of addressing selection bias and confounding, examine whether patients who follow the model's advice end up with better health outcomes than those who do not?

The common characteristic of these calibration channels is that they are anchored in causal chains independent of confirmation by the evaluated system itself. Physical experiments, court decisions, and long-term health outcomes can all provide external constraints, but their generation processes, task fit, and potential biases must

still be examined. Court records can serve as calibration material if isolated from the training data and matched to the task context; agreement with existing decisions is only one form of evidence. Long-term health outcomes are formed jointly by physiological processes and human actions, so treatment choices, adherence, selection bias, and confounding must still be addressed. Independence means that evidence is not confirmed by the evaluated system itself, not that it requires no human judgment or is unaffected by human action. These are **real windows**, not screens.

§3.4 Diagnostic Increment

What the engineering community sees is "when models become too strong, labelers cannot keep up." The Fiber Framework additionally asks: **is the labeler's judgment benchmark independent of the system being evaluated?** This problem depends not on model capability but on whether AI outputs actually enter the basis of judgment. Rising penetration of AI outputs may weaken labeler independence, but the relation is moderated by tasks, access to source material, and labeling procedures; neither strict monotonicity nor a universal threshold can be assumed. Only when feedback no longer supplies independent information does it cease to be independent calibration and become echo.

The value of this diagnostic increment is that it makes labelers' AI exposure an explicit monitoring variable. Model performance, inter-labeler consistency, and reward-model accuracy cannot by themselves establish whether labeler judgment is independent. The frequency, depth, task relevance, and source diversity of exposure must be examined separately when assessing RLHF's long-term calibration value.

Operational advice for practitioners: when designing an alignment pipeline, track labelers' AI exposure and its relation to the current task. If monitoring shows that feedback no longer supplies independent information, those data should no longer serve alone as independent calibration evidence. If retained, their dependencies must be documented and supplemented by independent calibration channels. Candidate supplements include labelers with low AI exposure and direct access to source material, historical human judgments checked for task relevance and temporal validity, and physical or outcome-based channels that do not depend on online preference. In the equation of §5, this mechanism supports only the possibility that the training-side supply of the AI narrative-load reduction coefficient (F_{AI}) is impaired; it does not imply that the unreduced proportion ($1 - F_{AI}$) must remain close to 1.

§4 The Third Increment: Taking Apart the Clock and Using a New Mapping to Approximate the Old Mapping Are Two Different Things

§4.1 Two Paths of Interpretability

There are two fundamentally different paths in contemporary interpretability research, and engineering literature usually subsumes them under the same title—"explainable artificial intelligence (XAI)." The Fiber Framework reveals that their epistemological status is entirely different.

Post-hoc attribution. Methods such as SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) and LIME (Local Interpretable Model-agnostic Explanations; Ribeiro et al., 2016) observe the input-output relation from outside the model and use a simple, interpretable proxy model to approximate the behavior of the original model in the neighborhood of a specific input. SHAP computes the marginal contribution of each feature on

the basis of Shapley values. LIME generates perturbed samples near the prediction point and trains a local interpretable model.

In the language of the clock room: the analyst does not take apart the clock—another, simpler clock is placed outside the clock, in the hope that the reading of the simple clock will reveal the working principle of the complex clock.

In the language of fiber, this is **a mapping of a mapping**. Through the original mapping function π_1, the original model maps the true state T into a reading r_1 ($\pi_1 : T \to r_1$). The attribution method works on the model, the input, and background conventions to produce feature-importance scores—between the explanation the analyst finally sees (a bar chart of SHAP values) and the true state T lies the attribution tool's own mapping. This layer of mapping has its own fiber. If the explanation merely recodes existing readings, it does not automatically remove the original information deficit concerning the true state T; but attribution tools also draw on the model and the input, so one cannot conclude from "another layer of explanation" alone that the fiber concerning the true state necessarily expands—one must check what evidence it adds and what conventions it relies on.

Post-hoc attribution changes both the behavioral evidence available to the analyst and the structure of error. Additional queries of the model and its inputs may reduce uncertainty about local input–output behavior, while the proxy approximation introduces its own error. The net effect depends on the added evidence, proxy fidelity, and background conventions. If an explanation merely recodes existing readings, it does not remove the original information deficit concerning the true state T; even an accurate account of model behavior does not establish external calibration to the true state T.

Mechanistic interpretability. The work of Elhage et al. (2022) and Conmy et al. (2023) proceeds along a completely different path. They directly inspect the model's internal representations—which neurons activate under a given input, how information flows across layers, and what kinds of "features" are encoded by the model's internal state. Elhage et al.'s superposition hypothesis tries to understand how the model compresses far more features than dimensions into a finite-dimensional representational space. Conmy et al.'s Automated Circuit Discovery tries to locate the subnetworks responsible for specific behaviors.

In the language of the clock room: the analyst removes the back cover of the clock and directly looks at the meshing of the gears.

In fiber language, mechanistic interpretability does not replace the original model with another input–output proxy; it obtains evidence about the internal representations and causal pathways of the original mapping function π_1. "Adding no proxy layer" does not mean that no observational mapping is present: reading, visualizing, and naming internal states remain dimensionality-reduced representations, and third-layer irreversibility still applies. Mechanistic evidence can reduce uncertainty about local internal mechanisms, but it does not automatically shrink the fiber concerning the true state T, nor does it guarantee less error than post-hoc attribution.

The distinction between mechanistic interpretability and post-hoc attribution can be given a deeper formulation through von Foerster's terminology. Von Foerster (2003) distinguishes between a "trivial machine," whose internal states do not change with the input, so that its input-output relation is fixed and predictable, and a "non-trivial machine," whose internal states evolve with the history of inputs, so that the machine becomes a different trivial machine, and remains unpredictable to any observer who does not know its internal states.

Von Foerster's distinction concerns historical state and input–output relations, whereas this book's meta-audit

criterion concerns independent reference; these are not the same classification. This book calls the process by which a subject examines the grounds of its own judgment and, when needed, introduces evidence outside the current narrative loop a meta-audit loop. This is a functional description of meta-audit capacity in individual cognition. It does not presuppose a dedicated module separate from existing cognitive processes, nor does it require proof that human cognition cannot be assigned to System 1 or System 2.

Whether current large language models possess the non-trivial component defined above must be examined through the state criterion, not inferred merely from the absence of an independent monitoring layer[2]. For a single forward computation with fixed weights and a fixed context, the internal transformation of that computation can be analyzed as a deterministic mapping; stochastic decoding must also be specified when present. Deployments retaining dialogue history, caches, or external memory require an account of how state updates alter subsequent behavior. Current architectural observations justify neither classifying LLMs wholesale as trivial machines nor inferring that they possess a meta-audit loop. What this book can assert is that "reflective" text produced within the same forward pipeline does not automatically constitute independent meta-audit across mapping levels.

This distinction gives mechanistic interpretability a more precise function: it supplies second-order observational material about a system's internal causal pathways from outside, without thereby installing a meta-audit loop in the system itself. Post-hoc attribution can also supply external second-order observational material, but its evidence comes chiefly from input–output relations and must not be presented as internal mechanistic evidence. The difference lies in the source of evidence, not in whether either method may be called second-order observation.

§4.2 The Operational Consequences of the Distinction

This distinction has not been clearly formulated in the current XAI literature through the Fiber Framework. Its operational consequences concern the respective boundaries of application and evidential strength of the two methods.

The correct expectation for post-hoc attribution methods. SHAP values describe how a model output is allocated among feature contributions under specified feature functions, background distributions, and attribution conventions. This is useful—it can generate the hypothesis that the model may rely on a feature. Contribution allocation within those conventions, however, is not identical to unique causal importance. Proxy fidelity can be examined with additional samples, perturbations, or interventions, while causal interpretation requires further identifying conditions. Approximation quality therefore is not unknowable in principle, but precision within a convention must not be mistaken for causal faithfulness.

Post-hoc attribution is suitable for **generating and screening hypotheses** and can supply local behavioral evidence under explicit operational definitions; it is not suitable for **verifying causal hypotheses on its own**. SHAP may indicate a high attribution weight for a sensitive attribute, but this result alone cannot prove discrimination. Causal and fairness judgments require additional evidence.

[2] In the three-layer architecture of "System A / System B / System M" proposed by Dupoux, LeCun, and Malik (2026), "System M" is a meta-control signal that monitors internal meta-states and organizes information flow between the learning components, switching between two learning modes (observational learning vs. behavioral learning). In this book's terms, it still routes within the visible subspace rather than crossing mapping levels; monitoring meta-states and switching learning modes are not sufficient by themselves to establish an independent calibration reference; the name "meta-control" alone therefore cannot register the process as this book's meta-audit loop.

The correct expectation for mechanistic interpretability. Taking apart the clock to look at the gears can verify specific causal hypotheses. When an experimenter activates or suppresses a group of neurons and observes the resulting change in model output, they are establishing a causal link—"when this group of neurons is activated, the model output changes in this particular way." For the same internal-pathway hypothesis, a suitably controlled internal intervention provides a more direct test than attribution alone. A change in output after intervention, however, does not by itself confirm the named semantic feature or its full role in the unperturbed state.

But mechanistic interpretability covers only the **local** behavior of the model—the analyst has seen how one set of gears meshes, but the clock has a trillion gears and they cannot be checked one by one. Even the most advanced current methods of circuit discovery can only localize the circuits behind relatively simple behaviors such as syntactic phenomena and factual recall. For complex reasoning behavior, a full circuit remains beyond our analytic capacity.

Local mechanistic findings cannot alone support overall risk assessment across tasks, populations, and deployment conditions; they must be combined with broader behavioral testing.

The complementarity of the two. Neither can replace the other: post-hoc attribution can provide an approximate behavioral map across a broad set of samples, while mechanistic interpretability can provide local causal evidence. The map cannot replace the rock sample, and the rock sample cannot replace the map.

§4.3 Diagnostic Increment

The engineering community usually puts both SHAP and mechanistic interpretability into the same "XAI toolbox" and lists their strengths and weaknesses under one label. The Fiber Framework reveals this: **the two stand at different epistemological levels**.

- Post-hoc attribution adds an explanatory mapping outside the original model. Additional model queries and input perturbations can supply new local behavioral evidence, while proxy and background-convention errors are also introduced; the net effect must be examined separately. It supplies no independent calibration to the true state T and therefore cannot turn behavioral explanation into an external calibration anchor.
- Mechanistic interpretability directly examines internal representations and causal pathways in the original model rather than replacing it with another input–output proxy. It can reduce uncertainty about local internal mechanisms, but observation and explanation retain their own mappings, and the fiber concerning the true state T does not automatically shrink merely because the model has been opened.

The value of this distinction is that it provides a principled way to judge, in a given scenario, which kind of explanation ought to be trusted and to what degree.

Operational advice for practitioners:

1. **Record evidential levels by the proposition being tested.** Attribution results can support contribution allocation under stated conventions or local behavioral descriptions; they cannot by themselves prove real-world causal effects or discrimination. Observing internal states can generate mechanistic hypotheses, while appropriate interventions and controls provide further evidence for particular causal pathways. Audit reports must specify data, conventions, interventions, and the scope of conclusions; credibility cannot be determined merely by the labels 'SHAP analysis' or 'mechanistic interpretability.'

2. **Investment and evidence allocation.** If system failure would cause irreversible harm, mechanistic evidence, causal interventions, and independent outcome validation deserve higher priority, but narrow mechanistic findings cannot replace broader behavioral audits. SHAP may indicate possible reliance on the wrong feature; confirming that reliance requires mechanistic interpretability, appropriate causal intervention, or other independent validation, not another attribution chart. In the equation of §5, this mechanism corresponds to actual narrative load (ϕ) being maintained or amplified: mistaking post-hoc attribution for sufficient validation may stop users from initiating further checks, but it does not imply unconstrained growth of the degeneration term.

§5 Conclusion: Manageable Imperfection

The three increments follow a single calibration chain: whether users retain conditions for active judgment, whether calibration sources still provide independent constraints, and whether the outputs of interpretability tools are tested appropriately. Engineering can support meta-audit not because a tool sits inside or outside the model, but because it incorporates independent evidence and preserves channels for review and correction. The Fiber Framework examines these conditions jointly, without exempting itself from the same examination.

The dynamical relation among the three conditions can be characterized by one equation. Let the user's meta-audit capacity be $A_{user}(t)$. Its rate of change is determined by two terms[3]:

$$\frac{dA_{user}}{dt} = -\alpha \cdot \Lambda \cdot \phi_0 \cdot (1 - F_{AI}) + \beta \cdot A_{user}^{intrinsic}$$

The first term is the degenerative contribution of AI interaction to the user's meta-audit capacity (A_{user}): the degeneration coefficient (α) modulates the strength of degeneration, and the media conductivity coefficient (Λ) is concretized here as the rate and precision with which AI outputs reach the user. This section denotes the narrative load in a specified reference deployment as reference narrative load (ϕ_0) and the load actually released as actual narrative load (ϕ), the latter following Chapter 3's definition. Within the same task, receiving boundary, and load scale, if $\phi_0 > 0$ and the engineering measures produce a net reduction, then $F_{AI} = 1 - \phi/\phi_0$ is the AI narrative-load reduction coefficient for that setting. Hence $\phi = \phi_0(1 - F_{AI})$. Here F_{AI} is an effect parameter relative to a specified baseline, not a general guardrail score and not the subject's meta-audit capacity. The interval $[0, 1]$ covers only net reduction. If a measure instead increases load, its negative effect must not be truncated to zero while preserving the equation; record the actual load directly or permit a signed effect parameter. The second term is the user's intrinsic recovery term: the meta-audit capacity baseline under specified reference conditions ($A_{user}^{intrinsic}$) is the level of capacity the user can maintain under those conditions, while the recovery rate (β) summarizes the availability and effectiveness of current recovery conditions. Offline activity, deep reading, and face-to-face contact count as such conditions only when they provide judgment practice, independent evidence, or relational work.

[3]This equation is a qualitative representation for a specified setting and a local time interval; it is not intended for long-run integration or terminal-state prediction. The degeneration coefficient (α) and recovery rate (β) cannot be quantified precisely, but the relative magnitudes of the two terms can be assessed directionally through proxies such as active verification, non-AI interaction time, and perception of uncertainty. Because the right-hand side does not explicitly contain the user's meta-audit capacity (A_{user}), it characterizes instantaneous drift rather than a closed-loop dynamic. The three increments enter through changes in reference narrative load (ϕ_0), the AI narrative-load reduction coefficient (F_{AI}), and recovery rate (β); this does not imply that they necessarily accelerate one another or that any parameter changes monotonically. The meta-audit capacity baseline under specified reference conditions ($A_{user}^{intrinsic}$) denotes the level of capacity the user can maintain under those conditions, not a stable equilibrium or a reserve that exists automatically without work.

The media conductivity coefficient (Λ) and the degeneration coefficient (α) are structural parameters: the media conductivity coefficient (Λ) is determined by the deployment ecology (the frequency and precision with which AI reaches the user in daily life), and the degeneration coefficient (α) by individual user differences (cognitive habits, degree of critical training); they are not directly driven by the three increments of this chapter; within the analytic horizon of this chapter they are treated as given parameters—over longer time scales, the media conductivity coefficient (Λ) also rises as user dependence deepens (Chapter 3, §2.2 defines the media conductivity coefficient as a technical variable shaped by power subjects, not an exogenous constant); this equation does not make that slow coupling explicit.

A nonnegative rate of change in the user's meta-audit capacity ($dA_{user}/dt \geq 0$) indicates no net decline during the interval examined; it is only a local criterion of capacity maintenance. Manageability also requires that current capacity be sufficient for the task and that independent evidence, review resources, and authority to correct actually be available. Under §2's conditions, the specified operationalization of HHH may keep reference narrative load (ϕ_0) high; under §3's conditions, training-side support for the AI narrative-load reduction coefficient (F_{AI}) may weaken; under §4's conditions, mistaken interpretation of explanatory evidence may leave actual narrative load (ϕ) uncorrected. These changes can increase degenerative pressure but do not individually imply monotonic growth of the degeneration term. Only independent evidence that reduced non-AI interaction lowers the recovery rate (β) supports a weakening recovery term. If the degeneration term remains larger than the recovery term over an interval, the user's meta-audit capacity (A_{user}) declines during that interval; the three conditions need not cross their boundaries simultaneously. This potentially self-reinforcing process is a testable engineering elaboration of Chapter 5, §3.2's claim that non-human objects manufacture demand for non-human objects.

- Examining each H only through its itemized metrics can miss their combined effect on user calibration. Asking whether the user's calibration behavior is declining makes the conditions for a narrative barrier available to inspection.
- Examining only labeler scores and reward-model optimization can miss the independence of the calibration source. Asking whether labeler judgment is independent of the evaluated system makes the self-reference loop available to inspection.
- Comparing the two methods only by the explanations they produce can miss their sources of evidence and structures of error. Asking what evidence and error each adds, and whether it directly examines causal structure, makes their epistemological difference identifiable.

The value of the Fiber Framework therefore does not lie in replacing engineering practice—it lies in providing a **structured diagnostic perspective**. The engineers are in the room repairing clocks, and the framework additionally asks: have these clocks been calibrated? Alignment. Has the calibrator itself been calibrated? The independence of the labeler. What evidence do these explanations provide, and how have their conclusions been tested? The sources and scope of explanatory evidence.

The final goal of engineering practice is not to reconstruct the complete true state. Chapter 1, §4.3 limits the recovery of all lost dimensions from lossy readings; it does not eliminate calibration or verification in particular tasks. Engineering can improve task accuracy while acknowledging that such improvement does not restore every dimension discarded by the mapping. The goal is **manageable imperfection**.

Where is the boundary of manageability? Each diagnostic increment marks a risk condition. Erosion of any one condition is insufficient by itself to establish unmanageability; that judgment requires persistent damage that independent calibration and recovery sources cannot offset:

1. **The condition from §2**: when the specified operationalization of HHH persistently reduces calibration behavior and no other calibration cues compensate, imperfection may become unmanageable.
2. **The condition from §3**: when calibrator independence declines and independent outcome-calibration channels cannot compensate, imperfection may become unmanageable.
3. **The condition from §4**: when explanatory error is mistaken for sufficient validation and other audits cannot identify that error, imperfection may become unmanageable.

The second layer of irreversibility in Chapter 1, §4.2, dynamic corrosion, mapping rewritten through use—the pressure of fiber expansion that calibration persistently fails to eliminate—is the dynamical basis of these conditions. Under the conditions listed in §2, the synergistic mechanism of HHH compresses the conditions for calibration; its net effect must still be judged in combination with the recovery term of this section. The loss of labeler independence means the drift of the calibration source itself. Misusing interpretability methods can conceal the explanation's own error; it does not necessarily increase uncertainty in every sense.

The three warning signs have now been erected. They do not demolish the room, and they do not eliminate mapping. They let the engineers inside the room know this: **engineers are repairing clocks, not repairing the sun. A repaired clock still needs a window.**

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Appendix A: Engineering Paradigms and the System Mapping of the Three Layers of Irreversibility

For the detailed fiber analysis of each engineering paradigm, see Chapter 11, §§4–§9. This appendix provides a condensed table marking the place of the three diagnostic increments in the main text within the overall

panorama. This table does not repeat the technical details of Chapter 11; it serves only to check each paradigm's object of measurement, evidential boundary, and correspondence to the chapter's three diagnostic increments.

In the table, "information deficit," "dynamic corrosion," and "ontological dimensional reduction" refer to the three layers of irreversibility argued in Chapter 1, §4: information deficit (when the mapping function is unknown, readings do not suffice to uniquely determine the true state), dynamic corrosion (the mapping is rewritten through use), and ontological dimensional reduction (when the true state is compressed into readings, dimensions are irreversibly lost).

Engineering paradigm	Core operation	Dilemma mainly addressed	Function in the language of fiber	Typical technologies/standards
Alignment	Fine-tune the model with human preference	information deficit + dynamic corrosion	Acquire partial knowledge of the mapping function; constrained by reward hacking (erosion by dynamic corrosion) and ontological dimensional reduction	RLHF, Constitutional AI, DPO
Guardrails	Set refusal boundaries for inputs and outputs	information deficit + dynamic corrosion (mainly); ontological dimensional reduction (acknowledgment)	Set refusal rules at the edges of known fibers; filter malicious input (information deficit), defend dynamically against adversarial attack (dynamic corrosion)	NeMo Guardrails, Llama Guard
Auditing	Systematically evaluate model behavior	information deficit (mainly); dynamic corrosion (drift detection)	Estimate overall risk and its distribution through behavioral indicators; detect performance decay over time	ISO/IEC 42001 (AI management system standard), NIST AI RMF
Red teaming	Simulate attackers to probe worst cases	information deficit (unknown blind spots); dynamic corrosion (adaptive vulnerabilities)	Seek the local extrema of fiber expansion—the input points where the model shifts abruptly from reliable to unreliable	Adversarial machine learning (ML), jailbreak testing

Engineering paradigm	Core operation	Dilemma mainly addressed	Function in the language of fiber	Typical technologies/standards
Robustness enhancement	Train the model to resist input perturbation	dynamic corrosion	Locally stabilize readings inside the neighborhood of input; faces an arms race of attacker adaptation	Adversarial training, data augmentation
Formal verification	Prove explicitly specified properties within a specified input region and under stated model assumptions	the boundary of ontological dimensional reduction	Partition "guaranteed zones" inside the fiber—in practice, verifiable properties are low-dimensional subsets of the behavior space	Reluplex (formal-verification solver for neural networks), deep neural network (DNN) verification
Interpretability (post-hoc attribution)	Attribution analysis such as SHAP/LIME	information deficit; ontological dimensional reduction (acknowledgment)	Provide local behavioral evidence through additional model queries and input perturbations while introducing proxy and background-convention error; do not independently constitute external calibration to reality; suitable for generating and screening hypotheses under specified conditions	SHAP, LIME

Engineering paradigm	Core operation	Dilemma mainly addressed	Function in the language of fiber	Typical technologies/standards
Interpretability (mechanistic)	Internal representation analysis, circuit discovery	information deficit (direct contact); ontological dimensional reduction (acknowledgment)	Directly examine internal representations and causal pathways to reduce uncertainty about local mechanisms; observation still involves mapping and does not automatically shrink the fiber concerning reality	Superposition analysis, Automated Circuit Discovery
Governance and compliance	Institutionalized evaluation and accountability	information deficit + dynamic corrosion + ontological dimensional reduction	Establish an accounting structure for tracking changes in fiber; institutional memory prevents forgetting	NIST AI RMF, EU AI Act
Continuous monitoring	Track performance shifts after deployment	dynamic corrosion	Keep looking at the sun only when independent outcomes, sampled human review, or other external feedback are incorporated; stable internal statistics alone do not constitute external calibration	Concept-drift detection, anomaly monitoring

Key judgment: no single paradigm can cover all three-layer dilemmas. Alignment addresses information deficit and part of dynamic corrosion, but is constrained by reward hacking and dimensional reduction. Guardrails acknowledge the boundary but do not solve calibration. Auditing estimates behavioral risk but cannot stop corrosion. Red teaming probes extremes but cannot structurally eliminate dimensional reduction. Governance provides bookkeeping but does not replace calibration at the technical layer. The engineering practice of Auditable AI must be a combined strategy—deploying complementary tools at different levels to form defense in depth.

Relation to the three increments in the main text:

- Main text §2, "HHH Is a Product," corresponds to the "Alignment" row in the table—marking the mech-

anism by which, under specific operationalizations, the three objectives jointly form a narrative barrier.

- Main text §3, “The Sun Becomes the Screen,” corresponds to the dynamic-corrosion dimension of the “Alignment” row—revealing the loss of independence of the calibration source itself.
- Main text §4, “Two Kinds of Interpretability,” corresponds to the two split rows in the table—revealing that beneath the unified label “interpretability” there hides a fundamental difference in epistemological level.